Low-Rank Multimodal Remote Sensing Object Detection With Frequency Filtering Experts

Xu Sun, Yinhui Yu, and Qing Cheng

***Abstract*— Visible-infrared object detection for remote sensing images plays an important role in the unmanned aerial vehicle (UAV) around-the-clock application. Most of the existing work focuses on designing complex network architectures to fuse complementary features, while few methods consider computational complexity and susceptibility against modality attacks, limiting the deployment of state-of-the-art frameworks. In this article, we present a low-rank multimodal object detection approach with frequency filtering experts, called LF-MDet, which is based on the advanced DINO (detection transformer with improved denoising anchor boxes) framework. This approach achieves more accurate detection with fewer computational resources. In particular, when the specific modality is attacked or missing, our method still maintains higher robustness toward such pervasive perturbations. Specifically, we propose a low-rank enhancement technology (LET) and a dynamic illumination-aware mask (DIM) module to enable a single backbone network in the form of batch formulation to unbiasedly and compatibly extract multimodal features. Furthermore, we design a lightweight frequency expert encoder (FEE) from the frequency domain perspective to efficiently fuse complementary features by filtering out amplitude noise components and mixing feature tokens. Extensive experiments are conducted on the multimodal remote sensing object detection datasets, VEDAI and DroneVehicle. The results demonstrate the superiority of the proposed approach over advanced multimodal remote sensing object detectors. Compared to the baseline method, our low-rank multimodal detector (LF-MDet) effectively reduces the floating point of operations (FLOPs) by approximately 65% while improving detection accuracy. The code is available at https://github.com/cq100/LF-MDet.**

***Index Terms*— Frequency features, low-rank adaptation (LoRA), object detection, remote sensing image.**

I. Introduction

With the increasing popularity of unmanned aerial vehicles (UAVs), object detection for remote sensing images has garnered significant attention in many fields, such as traffic surveillance, autonomous driving, and disaster relief [1], [2]. In recent years, deep neural networks (DNNs) have demonstrated remarkable success, particularly in object detection tasks [3]. Existing DNN-based detectors are primarily designed for the RGB modality that contains rich texture and color information. Under low illumination environments, their performance declines drastically, limiting UAV around-the-clock applications [4]. The infrared radiation (IR) image as an auxiliary modality can adapt to various light variations and weather changes but suffers from the drawback of few texture details [5]. As a result, the integration of RGB modality and IR modality benefits the accurate object detection for remote sensing images across a range of illumination conditions [6].

Recently, numerous multimodal remote sensing object detection methods have been proposed [7], [8]. Most efforts focus on designing cumbersome network structures to improve detection performance. For example, the multispectral detection transformer (DETR) [9] utilized two independent backbone networks to extract features for each modality and added a switch spectrum mechanism to eliminate bias extraction toward RGB modality features. Iterative cross-attention guided feature fusion (ICAFusion) [10] introduced an additional dual-modal feature fusion (DMFF) module based on the Transformer structure, which fuses the complementary information of RGB and IR modalities by modeling global features. These methods generally have expensive computational costs, which are difficult to deploy on resource-limited UAVs. Other work strives to design lightweight networks to tackle the challenges of computational complexity [11]. SuperYOLO [12] concatenated the RGB and IR images as input, and employed the detector with a single backbone network to accelerate the training and inference process. Such a fusion method fails to fully utilize multimodal features due to the lack of interaction and is susceptible to modality attacks. A robust multimodal object detection method for remote sensing images that can better balance detection accuracy and efficiency is still less explored, which deserves further research.

In our work, we attempt to efficiently extract and fuse the complementary features of RGB and IR modalities in the spatial and frequency domains, achieving competitive accuracy with limited computational resources. Our motivation stems from the batch formulation in RGB-D salient object detection [13], where two modalities are concatenated along the batch dimension so that a single backbone network can fast process dual-source data in parallel. When we directly apply the simple batch formulation to multimodal remote sensing object detection, the detection accuracy is poorer, as shown in Fig. 1. We argue the reason for the phenomenon is that a single backbone network has poor feature extraction capability for more complex remote sensing images due to limited parameters and is susceptible to modality preferences introduced by pre-training. In addition, there is a lack of efficient

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The authors are with the School of Communication Engineering, Jilin University, Changchun 130012, China (e-mail: yuyh@jlu.edu.cn; sunx22@mails.jlu.edu.cn; chengqing19@mails.jlu.edu.cn).

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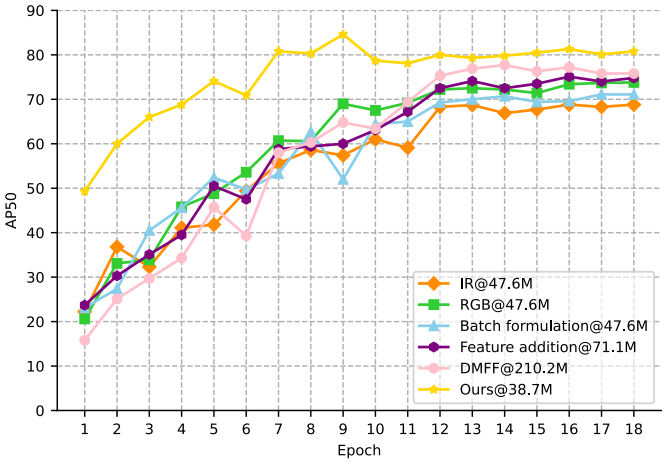


Fig. 1. Comparison of AP with 0.5 IoU threshold (AP50) on the VEDAI [14] dataset between our proposed method and some detection methods that are based on baseline DINO (DETR with improved denoising anchor boxes) [15]. Feature addition and DMFF [10] correspond to feature fusion operations for two backbone networks. The @ corresponds to the number of parameters.

and reasonable modality feature interactions to alleviate the learning pressure of a single backbone network.

Based on the above analysis, we propose an efficient visible-infrared remote sensing object detection approach based on the advanced DINO framework called low-rank multimodal detector (LF-MDet) with frequency filtering experts. In the spatial domain, we present a low-rank enhancement technology (LET) to enhance the compatible extraction ability of a single backbone network for multimodal features while preserving pre-training parameters for transfer learning. The technology performs a low-rank decomposition on some convolutional layers and integrates the decomposed matrices as bypasses. Meanwhile, we design a dynamic illumination-aware mask (DIM) module to prevent the model bias learning toward a particular modality. It takes the local light differences of RGB images as prior knowledge to selectively mask the corresponding part of each modality. In the frequency domain, we propose a frequency expert encoder (FEE) based on fast Fourier transforms (FFTs), which not only lightens the computing-intensive encoder in the DINO framework but also achieves efficient modality feature interactions. The proposed encoder filters out noise in the frequency amplitude components and adaptively mixes feature tokens from different modalities. Extensive experiments are conducted on the two RGB-IR remote sensing object detection datasets, DroneVehicle [16] and VEDAI [14]. Our main contributions are as follows.

1) We present a low-rank multimodal object detection approach with frequency filtering experts for remote sensing images called LF-MDet, which achieves higher detection accuracy with fewer computational resources. Our method also ensures robustness against attacks and missing perturbations on each modality.
2) We propose an LET and a DIM in the spatial domain to unbiasedly and efficiently extract complementary features from RGB and IR modalities.
3) We design an FEE from the frequency domain perspective to fuse and encode RGB and IR modality features through multilayer amplitude filters and frequency gate experts.

The remaining parts of this article are organized as follows. Section II discusses multimodal remote sensing object detection and frequency-domain deep learning. In Section III, the specific details of the proposed method are presented. Section IV describes the experiments and analyzes the results. Finally, a summary of our work is provided in Section V.

II. Related Work

A. RGB-Based Object Detection

Modern object detection methods based on deep learning can be simply categorized into the traditional CNN-based detectors and more recent DETR-based detectors [17], [18]. The convolutional models predict the locations and categories of objects by deploying manually generated anchor boxes or a grid of possible object centers, such as the anchor-based two-stage faster region-based convolutional neural network (Faster R-CNN) [19], one-stage you only look once version 4 (YOLOv4) [20], and the anchor-free fully convolutional one-stage (FCOS) object detection [21]. DETR-based models treat object detection as a set prediction task, establishing a direct mapping between the input image and the object set [22]. Many DETR-like variants primarily improve detector performance in terms of convergence speed and detection accuracy [23], [24]. Among them, the representative DINO [15] is an end-to-end Transformer detector that can greatly enhance training efficiency and small object detection capability.

Most of these detectors are based on RGB modality, which usually exhibits limited accuracy in low-light night or adverse weather conditions. This deficiency poses a challenge for UAVs operating in complex illumination scenarios. In this work, we develop an efficient multimodal object detection method based on the DINO framework for remote sensing images to enhance UAV around-the-clock detection capability. Even when the specific modality is attacked or missing, our method is still able to accurately detect objects.

B. Multimodal Remote Sensing Object Detection

RGB-IR remote sensing object detection focuses on effective extraction and sufficient fusion of complementary information to classify and locate objects [25], [26]. These methods can be roughly categorized into three types based on different fusion stages: pixel-level fusion, feature-level fusion, and decision-level fusion. In pixel-level fusion, multimodal images are simply concatenated or added as inputs to the detector [12], [27]. This approach usually requires complete feature alignment for RGB and IR modalities, especially for UAV remote sensing images where slight offsets for small objects lead to poorer detection accuracy. To address this issue, local adaptive illumination-driven input-level fusion (LAIIFusion) [28] designed a shift estimation module before the detector to calibrate object positions. Such a method is unable to load pre-training parameters, which may lead to poor robustness. Feature-level fusion usually uses two backbone networks to learn each modality features respectively,

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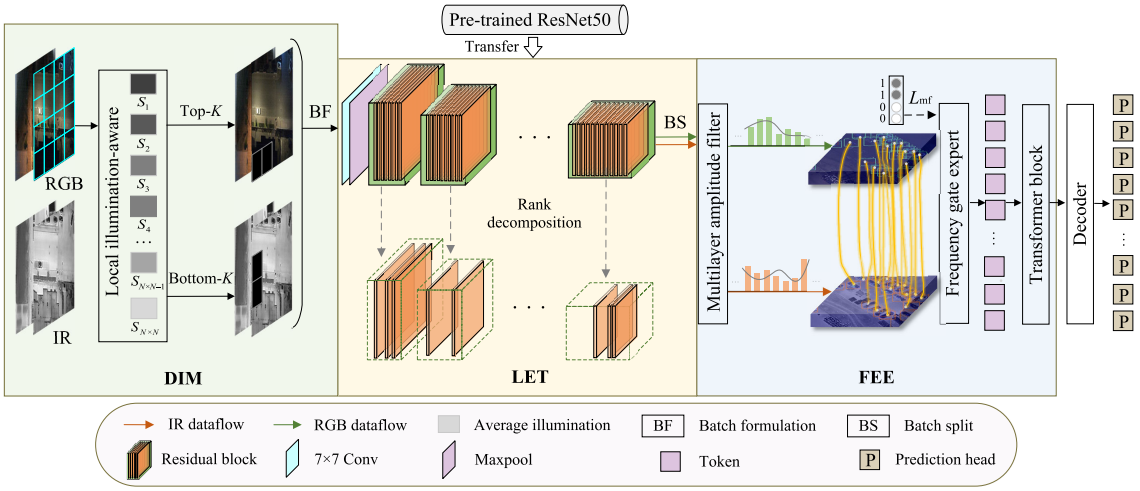


Fig. 2. Framework of the proposed LF-MDet. The green area is the DIM module, which masks parts of RGB and IR images. The yellow area is a single backbone network with LET. The blue area is the FEE, which consists of multilayer amplitude filters and frequency gate experts.

and subsequent fusion modules to mine complex integration relationships [29]. YOLOFusion [30] utilized a dual-source detection framework and added the cross-modality attentive feature fusion mechanism between two modalities to extract shared and independent features. Uncertainty-aware cross-modality vehicle detection (UA-CMDet) [16] generated uncertainty weights according to global light perception, and then assigned them to the RGB branch, IR branch, and fused modality branch. It ignored the variable lighting differences of local regions in the same image, limiting performance improvement. Cross-modality fusion transformer (CFT) [31] presented a cross-attention interaction module between two backbone networks to fuse the global contextual information. Although feature-level fusion is popular, dual backbone networks inevitably lead to higher deployment costs, which is difficult to apply on mobile devices, especially UAVs. Decision-level fusion independently supervises each modality and only contributes to the final decision [32]. It is a not common approach attributed to the drawback of completely ignoring the information interaction among modalities.

Our approach designs a single compatible backbone network in the form of batch formulation to unbiasedly extract multimodal features in parallel, which fully leverages pre-training and transfer learning to facilitate robust feature learning. Also, we take advantage of natural global features in the frequency domain to efficiently fuse different modality features, achieving a favorable trade-off between detection accuracy and efficiency.

C. Frequency-Domain Deep Learning

In recent years, deep learning has utilized frequency-domain methods from the traditional signal processing field to optimize DNNs [33], [34], such as learning non-local receptive field features [35] and enhancing domain generalization capability [36], [37]. Fast Fourier convolution [38] introduced a two-branch structure based on spectral transform theory, where frequency representation can be viewed as global features in the original domain to compensate for the deficiency of local features in vanilla convolutions. Deep frequency filtering [39] designed an instance-adaptive mask in the frequency domain to filter out frequency components adverse to domain generalization. It dynamically learned the spatial attention map and obtained the corresponding frequency information through element multiplication. Very recently, adaptive frequency filtering [40] has been used as an efficient global token mixer to replace large convolutional kernels. This primary neural operator can form a lightweight backbone network for downstream vision tasks.

At first glance, our proposed method may seem similar to the aforementioned frequency analysis approaches. However, our modeling perspective and network design differ substantially from them. We aim to leverage the Fourier Transform to achieve efficient extraction and sufficient fusion of multimodal complementary features. Furthermore, their structures are designed for a single modality and fail to consider the complex feature relationships in multimodal remote sensing images.

III. Proposed Method

A. Architecture Overview

The proposed LF-MDet follows the DINO framework, which is friendly to dense small object detection. Fig. 2 shows the overall architecture of our approach, which contains a DIM module, an LET, and an FEE. First, DIM selectively masks RGB and IR images by perceiving local illumination differences. These processed images are concatenated through the batch formulation as input. Then, LET transfers the parameters and structure of a single pre-trained ResNet50 [41], and injects low-rank decomposition weight matrices into some convolutional layers to compatibly extract multimodal features. These extracted features are split in batch concatenation order. Finally, FEE adopts multilayer amplitude filters to filter out the noise components for each modality in the frequency domain, and simultaneously learns global contextual information. The filtered features are fused by frequency gate experts and further

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encoded by the transformer block. In addition, we add an efficient modality fusion loss $\mathcal{L}_{\text{mf}}$ to supervise the feature fusion of FEE. The decoder and prediction heads are the same as DINO.

B. Spatial-Domain Feature Extraction

1) Dynamic Illumination-Aware Mask: In object detection based on DNNs, it is common practice to utilize pre-training that can speed up convergence and enhance model robustness [42], [43]. Currently, most pre-trained models employ RGB images due to their richer texture and color information. When directly transferring to multimodal detectors, the backbone network usually exhibits a bias toward extracting RGB modality features while ignoring the IR modality features. To this end, we propose a DIM module to avoid model bias learning, facilitating complementary feature extraction of a single backbone network. Unlike the random mask, our DIM combines local illumination-aware prior knowledge to simulate the model dependency on different modalities in real-world scenarios and reduces the gap between training and testing.

As shown in Fig. 2, parts of RGB and IR images are dynamically masked according to the local illumination, forcing the model to integrate information from the two modalities for object detection. The mask means that the pixel values are set to zero. Specifically, an RGB image is evenly divided into $N \times N$ patches. The average illumination S_i for each patch is calculated. These patches are sorted in ascending order of their illumination S_i. Afterward, we select the Top-K patches that are the darker regions, which may need to be enhanced with complementary information. The selected regions are masked on the corresponding locations of the RGB image. This manner enables the model to learn the features of these mask parts only from the IR modality, as it provides the missing information. Similarly, we select the Bottom-K patches that are stronger illumination in the RGB image, and subsequently mask these areas on the corresponding IR image locations. The discussion on the local illumination-aware prior knowledge, and hyperparameters N and K, is shown specifically in the ablation study of Section IV. In addition, we consider the case of complete modality loss. The missing modality is supplemented with a full mask image as input in the training phase.

2) Low-Rank Enhancement Technology: In order to use efficient batch formulations for parallel processing of multimodal data, it is desired to design a single backbone network with strong compatibility and the ability to fully utilize pre-training for robustness enhancement. We are inspired by the language model fine-tuning method known as low-rank adaptation (LoRA) [44]. It states that when pre-trained language models adapt to downstream tasks, the full-rank weight matrix usually has a lower intrinsic dimension that can efficiently learn features in smaller subspaces. In view of this, we propose an LET for the single backbone network. Unlike LoRA, our LET leverages low-rank weight matrices decomposed from some convolutional layers as bypasses to boost the complementary feature extraction capability and preserve pre-training. In the training phase, we completely fine-tune the single backbone network and the decomposed low-rank matrices without freezing any parameters.

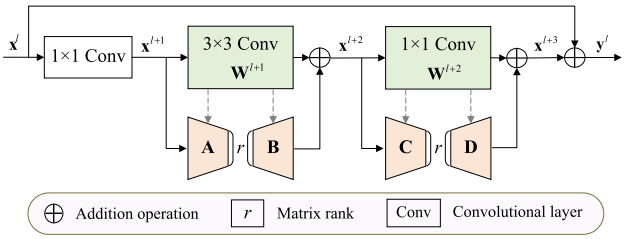


Fig. 3. Illustration of LET. In a residual block, the full-rank weight matrix $\mathbf{W}^{l+1}$ of the second convolutional layer is decomposed into low-rank matrices $\mathbf{A}$ and $\mathbf{B}$. The $\mathbf{W}^{l+2}$ of the last convolutional layer is decomposed into low-rank matrices $\mathbf{C}$ and $\mathbf{D}$.

Fig. 3 depicts the LET structure based on the pre-trained ResNet50 that consists of multiple residual blocks. Specifically, the RGB and IR image pairs processed by DIM are concatenated along the batch dimension as input, formulating the $(2b, h, w, 3)$ input size, where b stands for the batch size, h denotes the image height, and w represents the image width. In a residual block, the forward propagation process of the lth convolutional layer can be expressed as

$$\mathbf{x}^{l+1} = F\left(\mathbf{W}^l \mathbf{x}^l + \mathbf{b}^l\right) \tag{1}$$

where $\mathbf{x}^l$ is the input intermediate feature map, $\mathbf{x}^{l+1}$ is the corresponding output feature map, F represents the relu activation function, $\mathbf{W}^l$ is the lth convolutional layer weight matrix, and $\mathbf{b}^l$ is the bias. In our implementation, we decompose the full-rank weight matrices $\mathbf{W}^{l+1}$ and $\mathbf{W}^{l+2}$ for the last two convolutional layers of a residual block into low-rank matrices

$$\mathbf{W}^{l+1}_{m\times n} \rightarrow \mathbf{A}^{l+1}_{m\times r}\mathbf{B}^{l+1}_{r\times n} \tag{2}$$

$$\mathbf{W}^{l+2}_{c\times d} \rightarrow \mathbf{C}^{l+2}_{c\times r}\mathbf{D}^{l+2}_{r\times d} \tag{3}$$

where $\mathbf{A} \in \mathbb{R}^{m\times r}$, $\mathbf{B} \in \mathbb{R}^{r\times n}$, $\mathbf{C} \in \mathbb{R}^{c\times r}$, $\mathbf{D} \in \mathbb{R}^{r\times d}$, with rank $r \ll \min(m, n, c, d)$, $\rightarrow$ denotes the decomposition operation, and $\mathbf{AB} = 0$, $\mathbf{CD} = 0$ for initialization. The decomposed low-rank matrices as bypasses are multiplied with the same intermediate feature maps. The modified forward propagation process of a residual block is as follows:

$$\mathbf{x}^{l+2} = F\left(\mathbf{W}^{l+1}\mathbf{x}^{l+1} + \mathbf{ABx}^{l+1} + \mathbf{b}^{l+1}\right) \tag{4}$$

$$\mathbf{x}^{l+3} = F\left(\mathbf{W}^{l+2}\mathbf{x}^{l+2} + \mathbf{CDx}^{l+2} + \mathbf{b}^{l+2}\right) \tag{5}$$

$$\mathbf{y}^l = \mathbf{x}^{l+3} + \mathbf{x}^l \tag{6}$$

where $\mathbf{x}^{l+2}$ and $\mathbf{x}^{l+3}$ are the enhanced feature maps output from the last two convolutional layers, $\mathbf{b}^{l+1}$ and $\mathbf{b}^{l+2}$ are the corresponding biases, and $\mathbf{y}^l$ is the final output feature map of the enhanced residual block. We further explore the effect of matrix rank r and matrix decomposition position on performance in the ablation study of Section IV.

C. Frequency-Domain Feature Fusion

1) Frequency Expert Encoder: Current multimodal object detection methods usually design an independent and complex feature interaction module to fuse complementary information or model global contextual feature relationships [45], [46].

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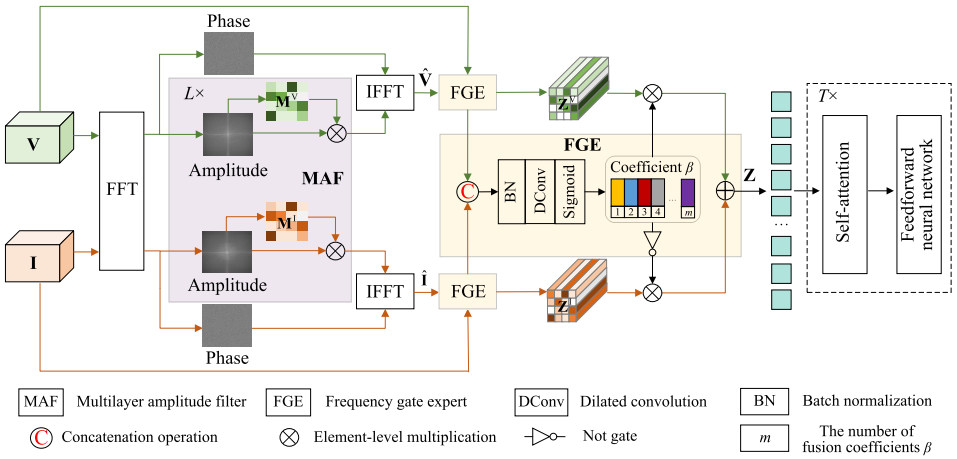


Fig. 4. Detailed architecture of the proposed FEE. The MAF filters the amplitude noise in the frequency domain for RGB and IR modality features, and then the FGE fuses the filtered intramodal and intermodal features.

Inevitably, this approach incurs additional computation costs. Also, the original encoder in the DINO framework is computationally heavy due to many stacked transformer blocks [47]. Therefore, we propose a lightweight FEE from the frequency domain perspective. The FEE combines the scene characteristics of multimodal object detection to lighten the original encoder. We exploit the advantage of the spectral transform theory that adjusting frequency components globally affects the spatial-domain features to efficiently filter and mix different modality features. The detailed FEE architecture is shown in Fig. 4, which mainly includes multilayer amplitude filters and frequency gate experts.

First, the features output from the backbone network are split into RGB modality feature map $\mathbf{V}$ and IR modality feature map $\mathbf{I}$ along the batch dimension. The two feature maps are transformed to the frequency domain by FFT

$$\mathbf{F}^{\mathrm{V}}(u,v)=\sum_{p=0}^{P-1}\sum_{q=0}^{Q-1}\mathbf{V}(p,q)\mathrm{e}^{-2\pi j(up+vq)} \tag{7}$$

$$\mathbf{F}^{\mathrm{I}}(k,l)=\sum_{p=0}^{P-1}\sum_{q=0}^{Q-1}\mathbf{I}(p,q)\mathrm{e}^{-2\pi j(kp+lq)} \tag{8}$$

where $\mathbf{F}^{\mathrm{V}}$ and $\mathbf{F}^{\mathrm{I}}$ are the transformed frequency feature maps, P and Q are the height and width of the $\mathbf{V}$ and $\mathbf{I}$, (p,q) is the position coordinate in the $\mathbf{V}$ and $\mathbf{I}$, u, v, k, and l are the frequency variables along the p and q directions, respectively. In the frequency domain, the $\mathbf{F}^{\mathrm{V}}$ and $\mathbf{F}^{\mathrm{I}}$ can be represented by amplitude and phase components. Taking $\mathbf{F}^{\mathrm{V}}$ as an example, the two components are expressed as follows:

$$\mathcal{A}^{\mathrm{V}}(u,v)=\sqrt{R^2\left(\mathbf{F}^{\mathrm{V}}(u,v)\right)+I^2\left(\mathbf{F}^{\mathrm{V}}(u,v)\right)} \tag{9}$$

$$\mathcal{P}^{\mathrm{V}}(u,v)=\arctan\left[\frac{I\left(\mathbf{F}^{\mathrm{V}}(u,v)\right)}{R\left(\mathbf{F}^{\mathrm{V}}(u,v)\right)}\right] \tag{10}$$

where $R(\mathbf{F}^{\mathrm{V}}(u,v))$ and $I(\mathbf{F}^{\mathrm{V}}(u,v))$ are the real and imaginary parts. In the same way, the amplitude $\mathcal{A}^{\mathrm{I}}$ and phase $\mathcal{P}^{\mathrm{I}}$ components of the $\mathbf{F}^{\mathrm{I}}$ can be obtained.

Then, we employ L multilayer amplitude filters to filter the noise in the amplitude components, as the amplitude component contains more low-level image information [48]. The RGB modality filter $\mathbf{M}^{\mathrm{V}}$ and IR modality filter $\mathbf{M}^{\mathrm{I}}$ are represented as follows:

$$\mathbf{M}^{\mathrm{V}}=\sigma\left[g\left(\mathrm{AvgPool}\left(\mathcal{A}^{\mathrm{V}}\right)\right),g\left(\mathrm{MaxPool}\left(\mathcal{A}^{\mathrm{V}}\right)\right)\right] \tag{11}$$

$$\mathbf{M}^{\mathrm{I}}=\sigma\left[g\left(\mathrm{AvgPool}\left(\mathcal{A}^{\mathrm{I}}\right)\right),g\left(\mathrm{MaxPool}\left(\mathcal{A}^{\mathrm{I}}\right)\right)\right] \tag{12}$$

where σ is the sigmoid activation function, [,] is a concatenation operation, AvgPool is the average pooling, and MaxPool is the max pooling. The g serves as a bottleneck layer, which includes a 1×1 Conv for feature dimension reduction, a relu activation function, and another 1×1 Conv for feature dimension increase. The filtered amplitude and original phase are used to reconstruct the RGB modality feature map $\widehat{\mathbf{V}}$ and IR modality feature map $\widehat{\mathbf{I}}$

$$\widehat{\mathbf{V}}=\mathcal{F}^{-1}\left(\mathbf{M}^{\mathrm{V}}\otimes\mathcal{A}^{\mathrm{V}},\mathcal{P}^{\mathrm{V}}\right) \tag{13}$$

$$\widehat{\mathbf{I}}=\mathcal{F}^{-1}\left(\mathbf{M}^{\mathrm{I}}\otimes\mathcal{A}^{\mathrm{I}},\mathcal{P}^{\mathrm{I}}\right) \tag{14}$$

where $\otimes$ is element-wise multiplication, and $\mathcal{F}^{-1}$ is inverse FFT.

Finally, we use the frequency gate experts to fuse the intramodal feature maps and intermodal feature maps. Mathematically, this process can be expressed as

$$\mathbf{Z}^{\mathrm{V}}=\Phi\left(\mathbf{V},\widehat{\mathbf{V}}\right) \tag{15}$$

$$\mathbf{Z}^{\mathrm{I}}=\Phi\left(\mathbf{I},\widehat{\mathbf{I}}\right) \tag{16}$$

$$\begin{aligned}\mathbf{Z}&=\Phi\left(\mathbf{Z}^{\mathrm{V}},\mathbf{Z}^{\mathrm{I}}\right)\\&=\beta\mathbf{Z}^{\mathrm{V}}+(1-\beta)\mathbf{Z}^{\mathrm{I}}\end{aligned} \tag{17}$$

$$\beta=\sigma\left(\mathrm{DConv}\left(\mathrm{BN}\left[\mathbf{Z}^{\mathrm{V}},\mathbf{Z}^{\mathrm{I}}\right]\right)\right) \tag{18}$$

where $\mathbf{Z}^{\mathrm{V}}$ is the fused feature map of the RGB branch, $\mathbf{Z}^{\mathrm{I}}$ is the fused feature map of the IR branch, $\mathbf{Z}$ is the final output fused feature map, Φ is the frequency gate expert function, β is the fusion coefficient, BN is the batch normalization, and DConv is a dilated convolution. In this encoder, we also use T transformer blocks with deformable attention, which consist of self-attention mechanisms and feedforward neural networks to further encode fused features. The T and L are discussed in the ablation study of Section IV.

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2) Modality Fusion Loss: We introduce a classification loss to guide the optimization of the FEE, which facilitates the extraction and fusion of complementary features. This loss aims to discriminate whether the receptive field corresponding to the fused feature element contains the object parts. Supposing that the fused feature map $\mathbf{Z}$ output from FEE contains M elements, our adopted binary cross entropy loss function $\mathcal{L}_{\text{mf}}$ can be formulated as follows:

$$\mathcal{L}_{\text{mf}} = \frac{1}{M}\sum_{i=1}^{M} -\left(z_i \log\left(\widehat{z}_i\right) + (1 - z_i)\log\left(1 - \widehat{z}_i\right)\right) \tag{19}$$

where z_i is the ground-truth label of the ith element, and $\widehat{z}_i$ is the predicted object existence probability. This loss $\mathcal{L}_{\text{mf}}$ is linearly combined with the classification and regression losses

$$\mathcal{L} = -\lambda_1 \mathcal{L}_{\text{GIoU}}(\mathbf{b}, \widehat{\mathbf{b}}) + \lambda_2 \|\mathbf{b} - \widehat{\mathbf{b}}\|_1 + \lambda_3 \mathcal{L}_{\text{FL}}(P(\mathbf{c})) + \lambda_4 \mathcal{L}_{\text{mf}} \tag{20}$$

where $\mathcal{L}$ denotes the aggregated loss, $\mathcal{L}_{\text{GIoU}}$ is the generalized intersection over union (GIoU) loss function, $\|\,,\|_1$ is the l_1 loss function, $\mathcal{L}_{\text{FL}}$ is the focal loss function, $\mathbf{b}$ and $\widehat{\mathbf{b}}$ are the ground truth and predicted bounding boxes, respectively, $\mathbf{c}$ corresponds to the semantic category of $\mathbf{b}$, and $P(\mathbf{c})$ represents the predicted classification score. The balance factors λ_1, λ_2, and λ_3 are the same as in the original DINO [15] experiments, which are 2.0, 5.0, and 1.0, respectively. The balance factor λ_4 for the introduced classification loss $\mathcal{L}_{\text{mf}}$ is discussed in the ablation study of Section IV.

IV. Experimental Results and Analysis

A. Datasets

Our experiments are conducted on the DroneVehicle [16] and VEDAI [14] datasets.

DroneVehicle is a recently released large-scale RGB-IR remote sensing object detection dataset captured by UAVs, containing 28 439 visible-infrared image pairs. The image size is 840 × 712 in our experiments. The dataset considers various complex shooting scenes and includes five vehicle types: car, truck, freight car, bus, and van. We choose 17 990 image pairs as the training set, 1469 pairs as the validation set, and 8980 pairs as the test set.

VEDAI is cropped from a much larger Utah Automated Geographic Reference Center (AGRC) dataset captured by satellites. We use the 1024 × 1024 image size. The dataset provides more than 1200 RGB-IR remote sensing image pairs covering a variety of scenes such as countryside, urban areas, and mountainous regions. It annotates eight categories: car, pickup, camping, truck, other, tractor, boat, and van.

B. Implementation Details

We implement the proposed approach based on the MMDetection toolbox 3.1.0 [49]. The comparative method YOLOv8l [50] employs the MMYOLO 0.6.0 [51] framework and does not use the default data augmentation parameters. The specific experiment parameters of our LF-MDet are shown in Table I. For hyperparameters, the matrix rank r is set to 4, patch number N is set to 4, Top/Bottom K is set to 4, the number of multilayer amplitude filter L is set to 2, the number of Transformer block T is set to 1, and the balance factor λ_4 for the classification loss $\mathcal{L}_{\text{mf}}$ is set to 0.89. Experiments show that our network reaches the best results under these settings. We will specifically discuss the impact of hyperparameters on the detection performance in Section IV, which guides parameter choosing.

TABLE I
Relevant Experiment Parameters

Parameters	Values
Hardware devices	NVIDIA Tesla A40 GPU (48GB)
Deep learning framework	PyTorch
Batch size	4
Epoch	18
Optimizer	AdamW
Initial learning rate	1.0×10^{-4}
Weight decay	1.0×10^{-4}

Five evaluation metrics are employed to quantify our method. As for model accuracy, we select the average precision (AP) with the IoU threshold from 0.5 to 0.95, AP50 with 0.5 IoU threshold, and AP75 with 0.75 IoU threshold. As for model efficiency, the floating point of operations (FLOPs) and the number of parameters (Params.) are used.

C. Comparison Experiment

We compare the proposed method with various popular object detection models: Faster RCNN [19], RetinaNet [52], YOLOv4 [20], FCOS [21], and DINO [15]. The detection results on the VEDAI and DroneVehicle datasets are shown in Tables II and III, respectively. It can be seen that the advanced DINO reaches higher AP scores of 48.3% and 43.1% on two datasets among comparative detectors with RGB modality as input. This exhibits the superiority of DINO for dense small object detection in remote sensing images. Compared to this unimodal DINO, our multimodal approach increases AP scores by 2.8% and 8.2% at less computational costs. To further compare the multimodal object detection performance, we employ the common feature-level fusion method with two backbone networks for these detectors, which includes two types of feature fusion manners. One is the simple feature addition operation. The other is the DMFF [10], which is a recently released leading approach based on global feature aggregation. As can be observed, the addition method leads to unstable detection accuracy, possibly due to the introduction of additional noise. Integrating DMFF slightly improves AP scores, but it occupies more computational resources. Compared to the second-best DINO + DMFF model, the proposed LF-MDet achieves higher AP scores of 51.1% and 51.3%, while reducing the number of parameters by nearly 82% and the FLOPs by approximately 65% on the two datasets. This proves the efficiency of our method in modeling global contextual feature interaction and fusing complementary information from different modalities.

To further highlight the superiority of our method, we conduct experiments to compare the proposed LF-MDet with

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TABLE II

DETECTION RESULTS OF COMPARATIVE EXPERIMENTS ON THE VEDAI DATASET. SUBCATEGORY ACCURACY IS EVALUATED BY AP. BOLDING INDICATES THE BEST RESULTS

Method	Modality	car	pickup	camping	truck	other	tractor	boat	van	AP	AP50	Params.(M)	GFLOPs
Faster RCNN	RGB	54.6	53.7	60.9	51.3	19.8	38.3	43.2	55.7	47.2	78.6	41.1	211.3
RetinaNet	RGB	51.9	46.3	54.9	44.2	19.0	19.5	18.6	14.7	33.6	54.6	36.5	212.3
YOLOv4	RGB	51.2	43.7	53.3	35.1	14.9	27.1	41.4	37.2	38.0	72.2	52.5	167.1
FCOS	RGB	53.1	51.3	54.0	44.5	20.5	31.5	32.1	15.1	37.8	63.3	**32.1**	206.5
DINO	RGB	55.6	53.0	57.7	43.7	28.1	48.3	40.9	59.4	48.3	73.8	47.6	284.0
Faster RCNN	IR	53.4	54.7	60.6	44.3	19.2	24.6	**44.6**	51.6	44.1	75.5	41.1	211.3
RetinaNet	IR	48.9	42.2	50.5	39.4	15.2	6.9	6.4	13.2	27.8	44.3	36.5	212.3
YOLOv4	IR	50.8	44.3	52.5	30.3	11.8	27.9	31.1	32.1	35.1	68.8	52.5	167.1
FCOS	IR	51.5	49.3	53.0	45.9	10.2	14.5	24.4	17.5	33.3	56.2	**32.1**	206.5
DINO	IR	50.9	55.1	56.3	42.7	28.3	39.9	38.4	51.8	45.4	68.8	47.6	284.0
Faster RCNN	RGB+IR(Addition)	54.9	50.9	60.1	43.0	17.8	30.1	39.9	42.5	42.4	72.3	64.6	297.2
RetinaNet	RGB+IR(Addition)	49.2	39.3	42.6	29.4	21.7	16.5	7.3	12.7	27.3	45.5	60.0	298.2
YOLOv4	RGB+IR(Addition)	51.3	47.2	55.5	34.5	13.6	19.0	41.8	33.8	37.1	67.6	93.1	316.1
FCOS	RGB+IR(Addition)	52.2	50.3	55.3	43.8	9.7	20.1	27.7	29.2	36.1	63.1	55.6	292.4
DINO	RGB+IR(Addition)	55.1	54.6	59.7	43.6	32.2	46.7	40.5	57.5	48.7	74.8	71.1	369.9
Faster RCNN	RGB+IR(DMFF)	57.0	55.9	59.8	**51.9**	24.7	38.7	41.7	52.8	47.8	78.2	203.7	341.0
RetinaNet	RGB+IR(DMFF)	48.8	50.5	43.1	37.0	23.7	29.2	21.1	31.0	35.5	66.4	199.1	342.0
YOLOv4	RGB+IR(DMFF)	52.5	45.3	50.4	40.5	25.1	28.8	38.6	32.7	39.2	64.9	232.2	359.9
FCOS	RGB+IR(DMFF)	49.7	49.7	52.9	40.7	16.8	29.4	35.2	33.6	38.5	64.5	194.7	336.2
DINO	RGB+IR(DMFF)	55.1	55.5	58.7	43.8	32.7	48.1	41.2	59.3	49.3	75.8	210.2	413.7
LF-MDet (Ours)	RGB+IR	**58.3**	**56.4**	**62.1**	44.3	**34.6**	**50.1**	42.3	**60.5**	**51.1**	**80.8**	38.7	**145.0**

TABLE III

DETECTION RESULTS OF COMPARATIVE EXPERIMENTS ON THE DRONEVEHICLE DATASET. SUBCATEGORY ACCURACY IS EVALUATED BY AP. BOLDING INDICATES THE BEST RESULTS

Method	Modality	car	freight car	truck	bus	van	AP	AP50	Params.(M)	GFLOPs
Faster RCNN	RGB	51.8	21.0	33.8	56.2	19.2	36.4	58.0	41.1	133.6
RetinaNet	RGB	52.1	8.3	18.4	37.5	9.4	25.1	42.5	36.5	129.0
YOLOv4	RGB	53.4	20.5	22.3	55.6	13.5	33.1	52.2	52.5	86.9
FCOS	RGB	52.5	10.3	20.8	44.3	11.0	27.8	43.9	**32.1**	125.3
DINO	RGB	51.8	30.3	42.5	60.6	30.6	43.1	63.6	47.6	146.0
Faster RCNN	IR	56.5	28.9	35.2	63.3	22.3	41.3	61.9	41.1	133.6
RetinaNet	IR	57.5	16.3	23.2	56.2	12.5	33.1	51.9	36.5	129.0
YOLOv4	IR	54.9	27.8	32.3	60.9	21.3	39.5	60.8	52.5	86.9
FCOS	IR	53.4	14.1	28.8	52.2	14.6	32.6	51.0	**32.1**	125.3
DINO	IR	57.1	36.8	45.5	**66.5**	29.7	47.1	66.2	47.6	146.0
Faster RCNN	RGB+IR(Addition)	35.7	21.3	26.3	36.8	20.0	28.0	39.0	64.6	183.5
RetinaNet	RGB+IR(Addition)	55.1	12.9	16.4	49.0	6.2	27.9	45.6	60.0	178.9
YOLOv4	RGB+IR(Addition)	54.6	25.6	28.7	61.1	19.3	37.9	58.4	93.1	173.4
FCOS	RGB+IR(Addition)	51.9	16.8	18.5	48.4	8.4	28.8	49.1	55.6	175.2
DINO	RGB+IR(Addition)	56.1	36.7	47.9	63.5	32.6	47.3	66.5	71.1	195.9
Faster RCNN	RGB+IR(DMFF)	55.5	29.5	41.9	60.9	29.7	43.6	62.8	203.7	215.9
RetinaNet	RGB+IR(DMFF)	56.3	21.5	28.3	56.7	15.1	35.6	53.0	199.1	211.3
YOLOv4	RGB+IR(DMFF)	57.2	25.4	35.2	63.0	23.0	40.8	59.9	232.2	205.8
FCOS	RGB+IR(DMFF)	51.1	18.3	30.6	50.2	17.0	33.4	52.9	194.7	207.6
DINO	RGB+IR(DMFF)	**58.3**	38.9	47.4	63.5	36.1	48.8	68.4	210.2	228.3
LF-MDet (Ours)	RGB+IR	57.3	**43.4**	**52.3**	64.4	**39.6**	**51.3**	**71.8**	38.7	**77.7**

the advanced multimodal object detection methods on the VEDAI and DroneVehicle datasets. The comparison results are provided in Tables IV and V. The SuperYOLO, CMAFF, and YOLOFusion are based on YOLOv5s, which focuses on lightweight design at the expense of detection accuracy [64]. As shown in Table IV, our LF-MDet achieves higher AP scores than the three methods. Multispectral DETR based on the same series of frameworks obtains the second best, but its FLOPs are more than twice as high as ours. In Table V, our approach exhibits impressive accuracy gains compared to the methods that occupy similar computational resources as ours. The CFT has competitive detection accuracy, but its parameters are approximately five times as large as ours. Our method achieves similar detection accuracy to the recently released C^{2}Former on the DroneVehicle dataset, but our parameters and FLOPs are lower. Moreover, we compare the advanced unimodal method YOLOv8 using RGB images or IR images. As can be seen, the proposed LF-MDet still improves the AP scores

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TABLE IV

PERFORMANCE COMPARISON OF MULTIMODAL OBJECT DETECTION METHODS ON THE VEDAI DATASET. WE ADOPT AP50 AS THE EVALUATION METRIC FOR SUBCATEGORY ACCURACY. YOLOV8 IS BASED ON THE MMYOLO FRAMEWORK WITHOUT THE DEFAULT DATA AUGMENTATION PARAMETERS. BOLDING INDICATES THE BEST RESULTS. ∗ THE RESULTS ARE TESTED ON RGB IMAGES. ⋆ THE RESULTS ARE TESTED ON IR IMAGES. † THE METHOD IS BASED ON OUR BASELINE DINO

Method	car	pickup	camping	truck	other	tractor	boat	van	AP	AP50	Params.(M)	GFLOPs
SuperYOLO [12]	90.6	83.3	77.7	68.5	52.4	82.1	62.2	69.4	48.1	73.3	**7.0**	81.5
ZAQ [53]	88.0	85.8	70.5	79.5	45.2	88.1	67.8	84.0	46.5	76.1	-	-
AFD [54]	88.5	85.5	71.4	73.3	58.7	89.1	59.8	80.5	45.4	75.9	-	-
CMAFF [30]	88.2	81.7	72.2	84.9	53.6	68.4	71.0	53.6	46.3	74.8	24.0	**34.4**
YOLOFusion [30]	91.7	85.9	78.9	78.1	54.7	71.9	71.1	75.2	49.1	78.6	24.5	34.9
GHOST [55]	88.8	86.1	74.3	**86.9**	61.3	86.0	**71.7**	**87.3**	49.1	80.3	-	-
Multispectral DETR [9]	-	-	-	-	-	-	-	-	50.8	-	73.0	328.0
YOLOv8∗ [50]	88.6	85.9	78.2	71.7	60.2	78.2	58.4	71.3	48.0	74.1	43.6	211.0
YOLOv8⋆ [50]	87.5	84.1	78.1	72.5	52.4	66.3	53.7	86.3	46.4	72.6	43.6	211.0
ICAFusion [10]	83.7	77.6	69.9	48.6	45.7	71.3	45.9	67.0	38.6	63.7	120.2	311.5
C^2Former [56]	87.2	80.7	82.7	77.4	58.4	72.9	71.4	75.2	48.3	75.7	101.0	430.5
C^2Former† [56]	90.8	84.5	**89.3**	84.2	64.4	**91.2**	61.9	65.1	49.6	78.9	79.0	374.5
LF-MDet (Ours)	**92.1**	**87.3**	85.9	71.6	**67.1**	89.5	68.8	84.1	**51.1**	**80.8**	38.7	145.0

TABLE V

PERFORMANCE COMPARISON OF MULTIMODAL OBJECT DETECTION METHODS ON THE DRONEVEHICLE DATASET. WE ADOPT AP50 AS THE EVALUATION METRIC FOR SUBCATEGORY ACCURACY. YOLOV8 IS BASED ON THE MMYOLO FRAMEWORK WITHOUT THE DEFAULT DATA AUGMENTATION PARAMETERS. BOLDING INDICATES THE BEST RESULTS. ∗ THE RESULTS ARE TESTED ON RGB IMAGES. ⋆ THE RESULTS ARE TESTED ON IR IMAGES. † THE METHOD IS BASED ON OUR BASELINE DINO

Method	car	freight car	truck	bus	van	AP	AP50	Params.(M)	GFLOPs
CSL [57]	88.4	32.1	22.4	60.0	22.5	26.8	45.0	37.1	109.6
KLD [58]	**89.9**	35.5	39.7	87.0	33.7	34.6	57.2	41.7	156.4
Gliding Vertex [59]	**89.9**	49.1	47.0	88.1	44.4	35.0	63.0	41.1	107.2
S^2ANet [60]	89.8	44.6	45.2	88.8	39.4	35.3	61.5	38.6	93.0
R^3Det [61]	89.8	49.6	48.2	88.1	44.5	36.8	63.8	37.0	107.6
ViT-B+RVSA [62]	89.7	51.0	53.2	88.0	44.4	42.6	65.1	60.5	134.7
I^2MDet [63]	-	-	-	-	-	46.2	-	**31.9**	**48.9**
UA-CMDet [16]	87.5	46.8	60.7	87.1	38.0	-	64.0	138.7	-
CFT [31]	81.3	57.9	71.5	86.2	53.8	50.1	70.1	206.0	403.9
YOLOv8∗ [50]	75.9	52.4	63.5	84.9	47.2	43.4	64.8	43.6	119.7
YOLOv8⋆ [50]	79.4	50.1	67.2	87.9	49.3	46.8	66.8	43.6	119.7
ICAFusion [10]	81.6	33.3	56.0	85.7	31.8	38.5	57.7	120.2	180.0
C^2Former [56]	82.7	48.6	67.0	87.7	44.0	46.5	66.0	101.0	258.3
C^2Former† [56]	83.1	**60.5**	69.6	**88.9**	55.7	**51.6**	71.6	79.0	224.7
LF-MDet (Ours)	82.2	59.6	**73.6**	86.6	**57.0**	51.3	**71.8**	38.7	77.7

by 7.9% and 4.5% on the DroneVehicle dataset. These results show that our proposed method achieves a favorable balance between detection accuracy and computational complexity.

For a more intuitive showing of the detection performance for the proposed method, we choose five different methods and our LF-MDet for qualitative comparison. The output visualization results on the VEDAI and DroneVehicle datasets are shown in Figs. 5 and 6, respectively. It can be observed that the unimodal YOLOv8 performs worse in low-lighting scenes than the proposed multimodal method. Except for our LF-MDet, other multimodal methods usually miss or mistakenly localize some small objects. As depicted in Fig. 6, the nighttime RGB image contains some invisible dark areas, while the IR image has many easily confused ghost shadows. Under such complex scenes, our method still accurately recognizes inconspicuous and similar objects, indicating that the LF-MDet enables UAVs to perform all-weather object detection.

In Fig. 7, we evaluate the robustness of advanced multimodal detection methods in different lighting conditions. We employ the gamma correction [65] as an attack method to simulate a variety of challenging UAV work scenes by adjusting the RGB image illumination. When the parameter Gamma is less than 1, the image becomes brighter. When the parameter Gamma is greater than 1, the image becomes darker. It can be seen that our LF-MDet still has stronger resistance in the face of diverse light variations. Our approach replacing the DIM module with the random mask also achieves the second best. Furthermore, we compare the robustness of these methods under different attack types. In Table VI, the performance of other multimodal approaches degrades drastically, especially when the preferred RGB modality is completely missing. In contrast, the proposed LF-MDet demonstrates more notable defense capabilities against modality attacks, illustrating that our approach consistently ensures higher robustness in

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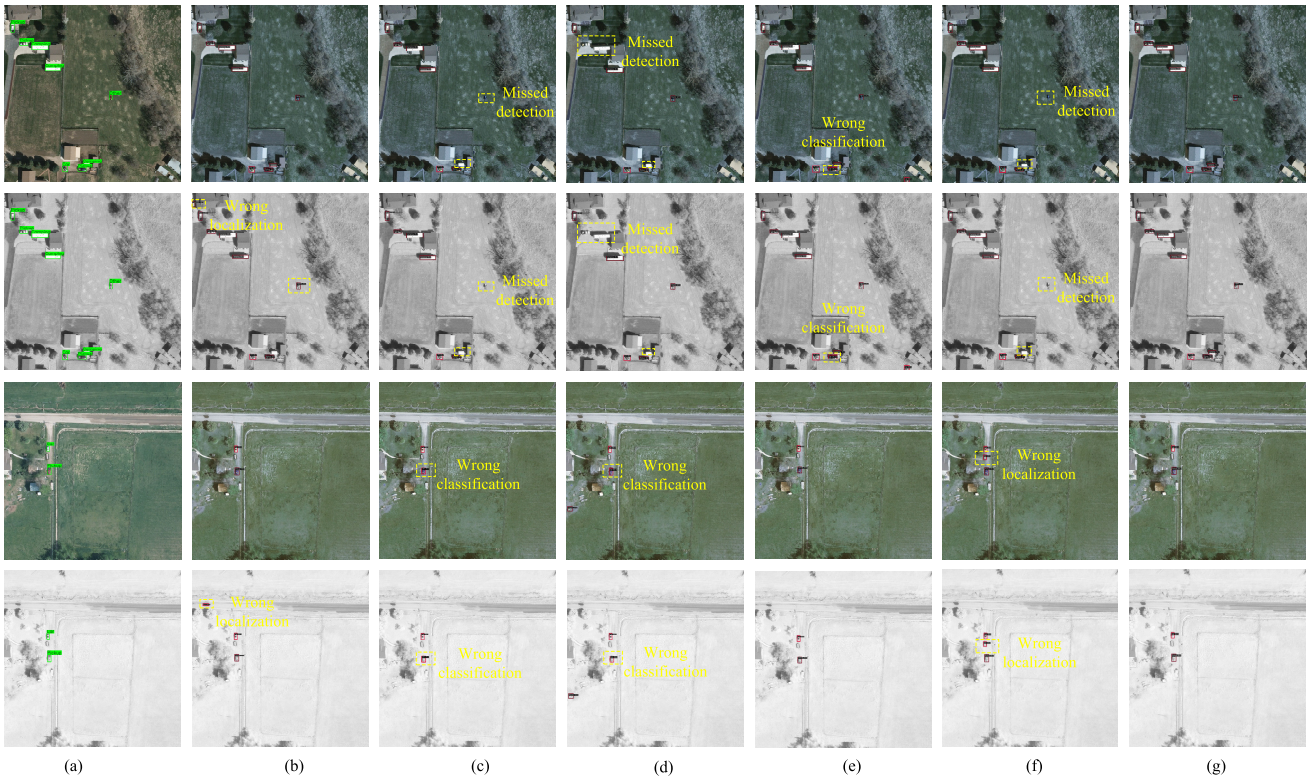


Fig. 5. Visual examples of different methods on the VEDAI dataset. (a) Ground-truth labels. (b) YOLOv8. (c) Original C^{2}Former. (d) Original ICAFusion. (e) C^{2}Former†. (f) ICAFusion†. (g) Our LF-MDet. The yellow dashed box indicates that the object has been missed or mistakenly localized. † The method is based on our baseline DINO.

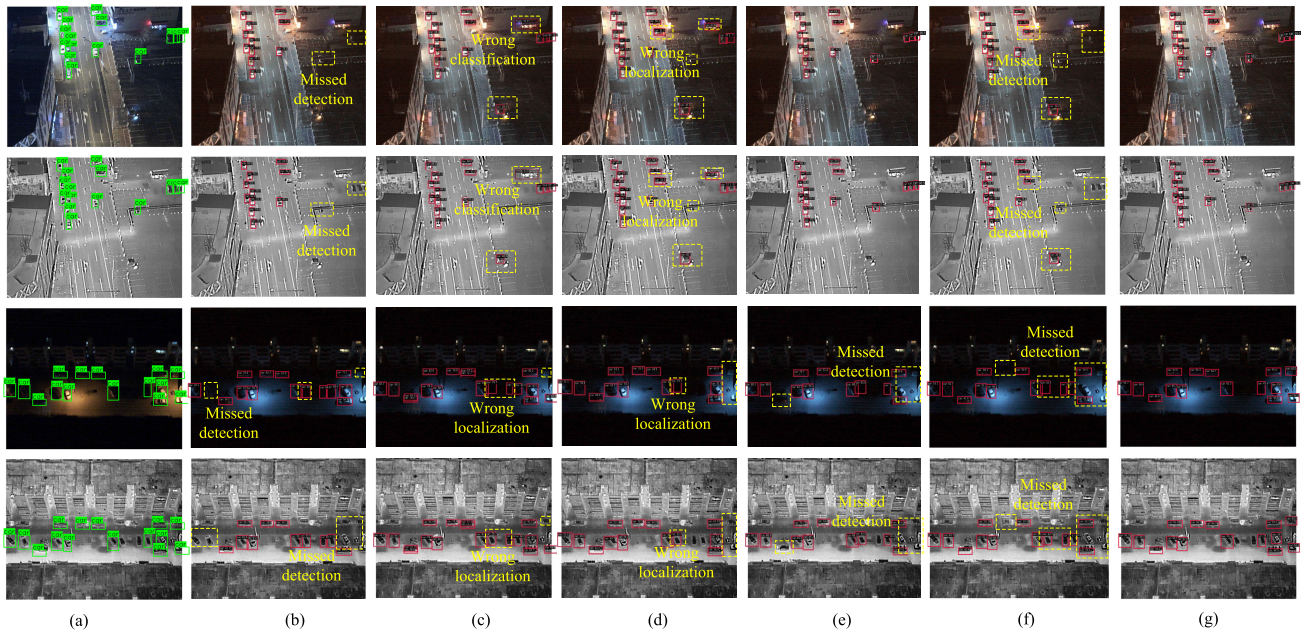


Fig. 6. Illustration of output visualization on the DroneVehicle dataset. (a) Ground truth labels. (b) YOLOv8. (c) Original C^{2}Former. (d) Original ICAFusion. (e) C^{2}Former†. (f) ICAFusion†. (g) Our LF-MDet. The yellow dashed box indicates that the object has been missed or mistakenly localized. † The method is based on our baseline DINO.

complex illumination scenes. This superiority stems from the fact that we can dynamically adjust model dependency on different modalities based on illumination-aware information, and develop a strong single backbone network to fully utilize the robust features retained from the pre-training.

D. Ablation Study

We perform some ablation studies on the VEDAI dataset to verify the effectiveness of the essential components in our LF-MDet, as shown in Table VII. This baseline is the DINO framework with ResNet50. The results show that directly using

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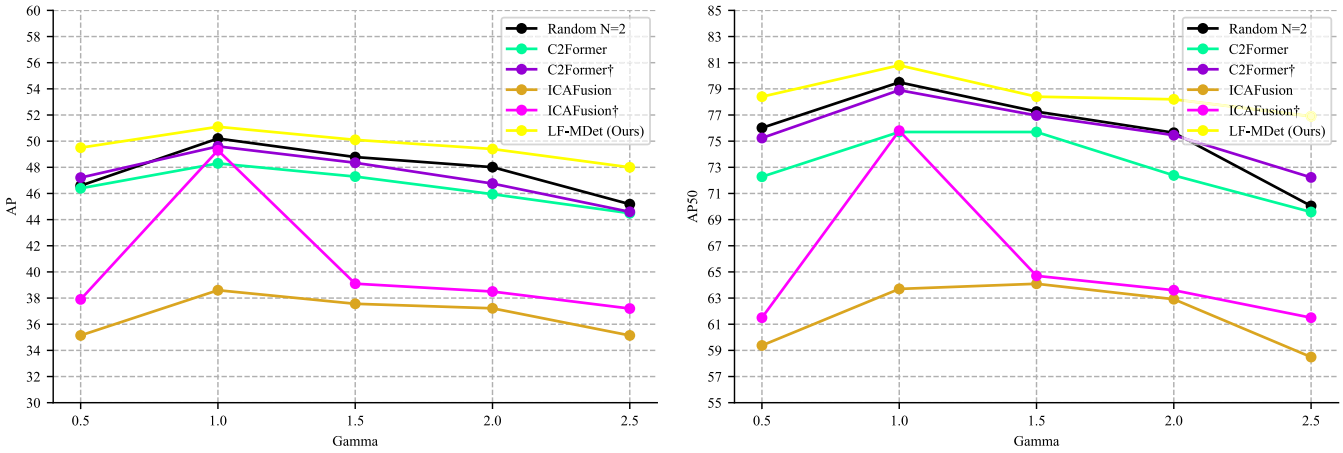


Fig. 7. Robustness evaluation of advanced multimodal detection methods in different lighting conditions on the VEDAI dataset. † The method is based on our baseline DINO. Random $N = 2$ indicates replacing the DIM module with the random mask in our approach.

TABLE VI

ROBUSTNESS COMPARISON OF ADVANCED MULTIMODAL DETECTION METHODS UNDER DIFFERENT ATTACK TYPES. BOLDING INDICATES THE BEST RESULTS. † THE METHOD IS BASED ON OUR BASELINE DINO. RANDOM $N = 2$ INDICATES REPLACING THE DIM MODULE WITH THE RANDOM MASK IN OUR APPROACH

Method	IR missing		RGB missing	
	AP	AP50	AP	AP50
Random N=2	37.1	64.1	20.7	42.3
C^2Former	26.5	46.0	6.5	19.5
C^2Former†	27.6	48.0	7.5	21.6
ICAFusion	14.4	24.2	4.5	8.7
ICAFusion†	25.5	45.4	5.0	9.0
LF-MDet (Ours)	**50.0**	**80.0**	**42.0**	**69.2**

batch formulation deteriorates detection performance. The proposed LET obtains AP scores comparable to the feature addition approach with similar computational complexity as the unimodal detection method. By integrating DIM, the AP score is improved by 1.4%, demonstrating that DIM facilitates multimodal feature extraction of a single backbone network. In addition, our FEE and modality fusion loss not only improve detection accuracy but also reduce the number of parameters and FLOPs. This is attributed to our method efficiently fusing multimodal features from the frequency domain perspective and lightening the computationally heavy encoder in the DINO.

To understand the contribution of LET in detail, we further explore the effect of different matrix ranks and matrix decomposition positions on the detection performance in Fig. 8. In particular, this experiment corresponds to the Baseline + LET in Table VII. The "1" indicates usage and "0" indicates non-usage. For example, "011" denotes the use of low-rank matrix decomposition in the last two convolutional layers of each residual block. As depicted in Fig. 8, when the rank r is 4, the detection accuracy is more stable than that of other values. Especially at position "011," the AP score reaches up to 48.4%. We also consider the computational costs and chose this setting. In addition, the curve for $r = 16$ seems to fluctuate obviously, but the performance degradation upper bound remains below 1.5, which falls within a reasonable range of performance fluctuation.

For the DIM, different patch numbers and Top/Bottom-K are further investigated in Table VIII. This ablation study corresponds to the Baseline + LET + DIM in Table VII. As can be seen, our DIM achieves higher AP scores compared to the random mask method, demonstrating the rationality and effectiveness of introducing local illumination-aware prior knowledge. When N is set to 4 and K is set to 4, the AP score reaches a maximum value of 49.8%. It indicates that our approach can better leverage the prior knowledge in this setting.

In Fig. 9, we explore the multilayer amplitude filter numbers L and Transformer block numbers T in the FEE. As can be seen, the performance at $L = 2$ outperforms that of most values. When L increases, the AP and AP50 scores decrease. We attribute this phenomenon to the fact that excessive filtering loses crucial information within each modality. Further, our method has the highest detection accuracy and the lowest computational complexity at $T = 1$. This suggests that using one Transformer block in our frequency expert mechanism can effectively learn mostly meaningful fusion features for multimodal remote sensing images.

We conduct the ablation study about the normalized weight α of the balance factor λ_4 in Table IX, where $\alpha = \lambda_4/(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4)$. The weight reflects the contribution degree of the introduced classification loss $\mathcal{L}_{\text{mf}}$ to the proposed method. It can be seen that our LF-MDet increases the AP score by 0.5% compared to the version without the classification loss ($\alpha = 0$). As the α progressively increases, the AP score initially rises, achieving a maximum value of 51.1%. Subsequently, the detection accuracy gradually decreases. The AP peak is reached at $\lambda_4 = 0.89$. These results indicate that the introduced classification loss can bring performance gains.

For a more intuitive demonstrating the advantages of frequency-domain feature fusion, we visualize some feature

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TABLE VII
RESULTS OF THE ABLATION STUDY FOR DIFFERENT ESSENTIAL COMPONENTS IN LF-MDET. BOLDING INDICATES THE BEST RESULTS

Method	AP	AP50	AP75	Params.(M)	GFLOPs
Baseline+RGB	48.3	73.8	50.9	47.6	284.0
Baseline+IR	45.4	68.8	49.2	47.6	284.0
Baseline+Batch formulation	46.8	71.1	51.8	47.6	284.0
Baseline+Feature addition	48.7	74.8	53.6	71.1	369.9
Baseline+DMFF	49.3	75.8	52.2	210.2	413.7
Baseline+LET	48.4	73.6	52.7	47.9	284.1
Baseline+LET+DIM	49.8	76.3	52.8	47.9	284.1
Baseline+LET+DIM+FEE(Our LF-MDet)	**51.1**	**80.8**	**57.9**	**38.7**	**145.0**

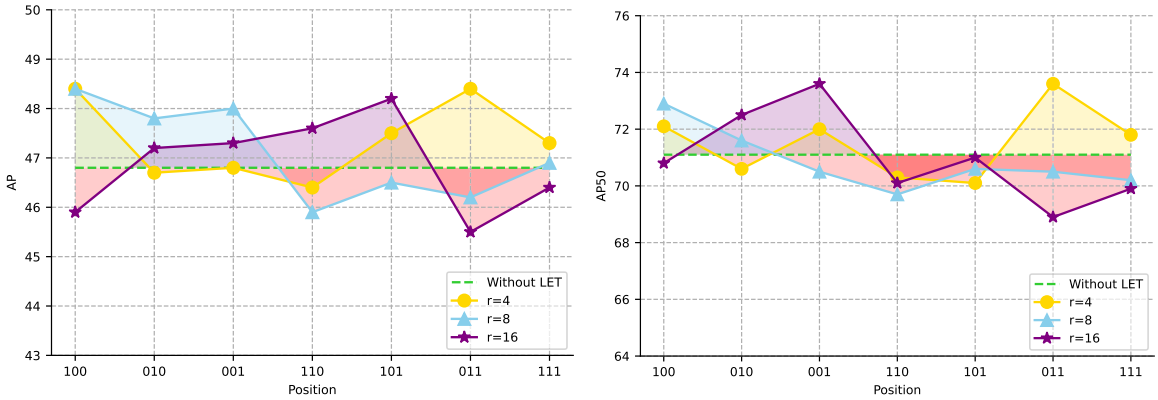


Fig. 8. Effect of different ranks r and matrix decomposition positions on detection performance. We adopt AP and AP50 as evaluation metrics. Without LET indicates that only baseline DINO and the batch formulation are used.

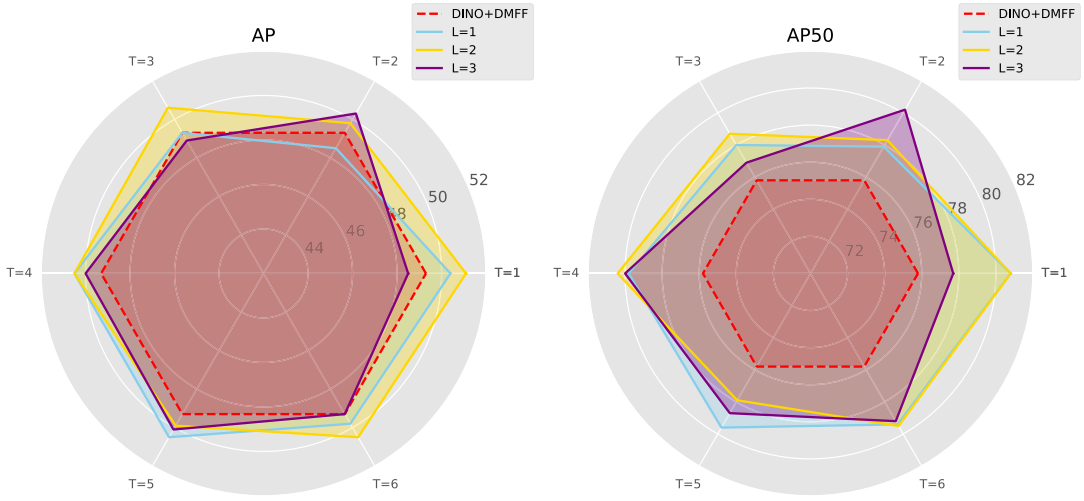


Fig. 9. Performance comparison of different multilayer amplitude filter numbers L and Transformer block numbers T in the FEE. We adopt AP and AP50 as evaluation metrics. DINO + DMFF indicates using the original DINO encoder.

maps of our LF-MDet and DINO + DMFF on the DroneVehicle dataset in Fig. 10. Our method includes the split feature maps from the LET output, the filtered feature maps through multilayer amplitude filters, and the fused feature map of the two modalities from the frequency gate expert output. The activated regions in the predicted bounding box represent the dependency relationship between decision behavior and the corresponding layer features. It can be observed that the LET has captured most of the object features. After frequency-domain filtering, the dependence on unreliable features in dark regions of RGB images is weakened and the dependence on reliable features is strengthened, which reduces missed detection and wrong localization similar to DINO + DMFF. Furthermore, our frequency gate expert effectively integrates the complementary features of RGB and IR modalities. This indicates that frequency-domain feature fusion enables our LF-MDet to accurately detect objects by balancing the two modality contributions.

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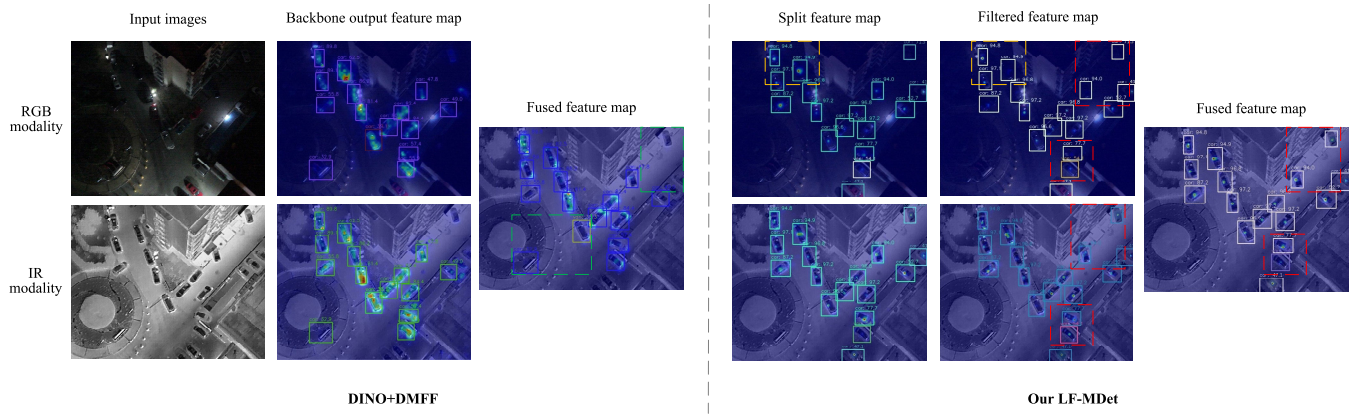


Fig. 10. Visualizations of some feature maps on the DroneVehicle dataset. The green dashed box indicates the wrong detection in the DINO + DMFF. In our LF-MDet, the yellow dashed box denotes the unreliable feature filtering. The red dashed box represents complementary feature fusion from RGB and IR modalities.

TABLE VIII

ABLATION RESULTS ON THE INFLUENCE OF THE PATCH NUMBER N AND TOP/BOTTOM-K. BOLDING INDICATES THE BEST RESULTS

$N \times N$ patches	Top/Bottom-K	AP	AP50	AP75
1×1	0	48.4	73.6	52.7
	1	49.5	76.1	55.4
2×2	2	47.8	72.9	52.7
	Random	49.2	74.8	54.3
	1	48.6	74.5	57.0
	2	48.5	75.5	54.9
3×3	3	48.2	73.0	56.5
	4	46.3	71.0	53.6
	Random	48.6	74.5	53.5
	1	47.4	72.4	55.0
	2	48.9	75.2	**57.8**
	3	48.4	72.8	54.6
	4	**49.8**	**76.3**	52.8
4×4	5	47.9	72.9	54.1
	6	47.7	73.1	55.9
	7	48.3	74.0	56.3
	8	47.5	71.8	56.2
	Random	48.2	74.2	53.2

TABLE IX

EVALUATION OF THE NORMALIZED WEIGHT α FOR THE BALANCE FACTOR λ_4 IN THE INTRODUCED CLASSIFICATION LOSS $\mathcal{L}_{\text{mf}}$. BOLDING INDICATES THE BEST RESULTS

Normalized weight α	AP	AP50	AP75
0	50.6	79.5	57.6
0.1	**51.1**	**80.8**	**57.9**
0.2	50.8	80.2	57.7
0.3	50.2	79.1	57.5

V. CONCLUSION

In this article, we propose a low-rank multimodal object detection approach with frequency filtering experts for remote sensing images, called LF-MDet, enabling the UAV around-the-clock operation. Our method shows a favorable trade-off between the detection accuracy and the computational complexity. The main components of the LF-MDet are an LET, a DIM module, and an FEE. The LET boosts the multimodal feature extraction ability of a single backbone network by low-rank matrix decomposition and preserves pre-training for robustness enhancement. The DIM addresses the challenge of model bias learning toward a particular modality and favors our LF-MDet to resist the pervasive perturbation of modality attacks. The FEE efficiently filters and fuses multimodal features from the frequency domain perspective while lightening the computationally heavy encoder in the DINO framework. The qualitative and quantitative experiments on two RGB-IR remote sensing object detection datasets demonstrate that our approach outperforms advanced multimodal object detectors. In addition, this efficient design can reduce the FLOPs by approximately 65% while improving detection accuracy over the baseline method.

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Xu Sun received the M.S. degree from the College of Communication Engineering from Jilin University, Changchun, China, in 2022, where she is currently pursuing the Ph.D. degree.

Her research interests include artificial intelligence and remote sensing.



Yinhui Yu received the Ph.D. degree from Jilin University, Changchun, China, in 2005.

She is currently a Professor with the College of Communication Engineering, Jilin University. Her research interests include the Internet of Things and artificial intelligence.



Qing Cheng received the M.S. degree from the College of Communication Engineering from Jilin University, Changchun, China, in 2022.

His research interests include computer vision and artificial intelligence.

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